

Introduction

As young adults, our focus naturally starts shifting towards becoming independent individuals, including the goal of living on our own and eventually becoming homeowners to complete part of the American Dream. However after experiencing high rent firsthand, this goal then starts to feel somewhat impossible. Housing affordability seems to be an increasingly urgent issue, especially in Charlotte, North Carolina. This then sparked our interest in figuring out and identifying the key factors that most contribute to housing unaffordability and then use that to identify the neighborhoods most at risk of displacement. The topic is even more so interesting due to the rising housing costs disproportionately affecting lower-income families and historically marginalized communities. This frequently forces residents out their homes and enforces the history of discriminatory practices such as redlining mapping. Redlining was used to mark hazardous neighborhoods in terms of investment which usually related to minority based neighborhoods. This raises questions about whether old policies continue to frame present-day outcomes.

When curiosity sparked around this topic, brainstorming different classification and regression models took place. After much refining and model training, our team ended up with an XGBoost model as our best performing model for the classification modeling and a generalized additive model of four features as our best performing model for the regression model. Throughout this report, we will break down the journey of deciding on the selection of these two models and how we managed to intertwine both of these models into reaching our analysis of the presence of impact of historical redlining on present day housing cost burden and displacement risk.

